

Lecture 14: Backward Stability; Vector and Matrix Norms

Math 6610

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Today's Goals

- Prove the accuracy theorem $\|\tilde{y} - y\| / \|y\| = O(\kappa \epsilon_{\text{mach}})$
- Review vector norms $(1, 2, \infty)$ and the Frobenius norm
- Define induced (lub) matrix norms and compute the classical three
- Prove submultiplicativity $\text{lub}(AB) \leq \text{lub}(A) \text{lub}(B)$

Accuracy Theorem

Remark (Backward stable (Lecture 13)). The computed answer is the exact answer for a slightly changed input: for each $x \neq 0$ there is $\tilde{x}$ with

$$\tilde{F}(x) = F(\tilde{x}), \quad \|\tilde{x} - x\| \leq C \epsilon_{\text{mach}} \|x\| \quad \text{for all small } \epsilon_{\text{mach}},$$

with C the same for every x .

Remark (Condition number (Lecture 9)). For $\delta > 0$ let $g(\delta)$ be the worst amplification over all input changes δx with $0 < \|\delta x\| \leq \delta$:

$$g(\delta) := \sup_{0 < \|\delta x\| \leq \delta} \frac{\|F(x + \delta x) - F(x)\| / \|F(x)\|}{\|\delta x\| / \|x\|}, \quad \kappa(x) := \limsup_{\delta \rightarrow 0} g(\delta).$$

(“sup” = the largest value, or the limit of the values if no largest one exists.) In words: relative change out $\div$ relative change in, for the worst small change.

Accuracy of a backward stable algorithm

Theorem (Accuracy of backward stable algorithms). Let $\tilde{F} : X \rightarrow Y$ be a backward stable algorithm for $F : X \rightarrow Y$. Let $x \in X$ with $x \neq 0$, $F(x) \neq 0$, and $0 < \kappa(x) < \infty$. Then there is $\epsilon_0 > 0$ (depending on x) such that

$$\frac{\|\tilde{F}(x) - F(x)\|}{\|F(x)\|} \leq 2C \kappa(x) \epsilon_{\text{mach}} \quad \text{whenever } \epsilon_{\text{mach}} \leq \epsilon_0,$$

with C from backward stability. In short:

$$\|\tilde{F}(x) - F(x)\| / \|F(x)\| = O(\kappa(x) \epsilon_{\text{mach}}).$$

Remark. The factor 2 can be replaced by any number larger than 1 (take ϵ_0 smaller).

Accuracy Theorem: Some Intuitions

Remark.

- 1 The algorithm hands back $F(\tilde{x})$: the exact answer for an input that is wrong by a relative amount of at most $C \epsilon_{\text{mach}}$ (backward stability).
- 2 The problem turns a small relative input error into a relative output error at most about $\kappa(x)$ times larger (conditioning).

So

$$\underbrace{\frac{\|\tilde{F}(x) - F(x)\|}{\|F(x)\|}}_{\text{output error}} = \underbrace{\frac{\text{output error}}{\text{input error}}}_{\lesssim \kappa(x)} \times \underbrace{\frac{\|\tilde{x} - x\|}{\|x\|}}_{\text{input error} \leq C \epsilon_{\text{mach}}} \lesssim \kappa(x) \cdot C \epsilon_{\text{mach}}.$$

Nothing else about the algorithm is used: every backward stable algorithm gets this bound for free.

Remark. “About $\kappa(x)$ ” holds only for small input changes. The proof makes this precise: for small ϵ_{mach} , “about $\kappa(x)$ ” becomes “at most $2\kappa(x)$ ”.

Accuracy Theorem: Proof

Proof. Idea: error = (amplification) $\times$ (input error) $\leq 2\kappa(x) \cdot C \epsilon_{\text{mach}}$.

1. *Error as an input change.* Backward stability gives $\tilde{x} = x + \delta x$ with

$$\tilde{F}(x) = F(x + \delta x), \quad \frac{\|\delta x\|}{\|x\|} \leq C \epsilon_{\text{mach}}.$$

(If $\delta x = 0$ the error is 0; so let $\delta x \neq 0$.)

2. *Split into two factors.* Multiply and divide by $\|\delta x\| / \|x\|$:

$$\frac{\|\tilde{F}(x) - F(x)\|}{\|F(x)\|} = \underbrace{\frac{\|F(x + \delta x) - F(x)\| / \|F(x)\|}{\|\delta x\| / \|x\|}}_{\text{amplification}} \cdot \underbrace{\frac{\|\delta x\|}{\|x\|}}_{\leq C \epsilon_{\text{mach}}}.$$

(continued on the next slide)

Accuracy Theorem: Proof (Cont'd)

Proof (continued). 3. *Amplification* $\leq 2\kappa(x)$. The worst amplification $g(\delta)$ only shrinks as $\delta \rightarrow 0$, down to $\kappa(x) > 0$. So there is $\delta_0 > 0$ with

$$g(\delta) \leq 2\kappa(x) \quad \text{for } 0 < \delta \leq \delta_0.$$

Our change has $\|\delta x\| \leq C \epsilon_{\text{mach}} \|x\| \leq \delta_0$ once $\epsilon_{\text{mach}} \leq \epsilon_0 := \delta_0 / (C \|x\|)$.

4. *Combine*. Multiply the two bounds:

$$\frac{\|\tilde{F}(x) - F(x)\|}{\|F(x)\|} \leq 2\kappa(x) \cdot C \epsilon_{\text{mach}}.$$



Vector Norms

Definition (Norm). A norm on $\mathbb{C}^n$ is a function $\|\cdot\| : \mathbb{C}^n \rightarrow \mathbb{R}$ (a “length”) such that for all $x, y \in \mathbb{C}^n$, $\alpha \in \mathbb{C}$:

- ❶ $\|x\| \geq 0$, and $\|x\| = 0$ if and only if $x = 0$
- ❷ $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality)
- ❸ $\|\alpha x\| = |\alpha| \|x\|$

Example (The 1-, 2- and ∞ -norms).

$$\|x\|_1 = \sum_{i=1}^n |x_i|, \quad \|x\|_2 = \sqrt{\sum_{i=1}^n |x_i|^2}, \quad \|x\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

(Total size, Euclidean length, largest entry.)

Matrix Norms

Definition (Matrix norm). A matrix norm is a function $\|\cdot\| : \mathbb{C}^{n \times n} \rightarrow \mathbb{R}$ such that for all $A, B \in \mathbb{C}^{n \times n}$, $\alpha \in \mathbb{C}$:

- 1 $\|A\| \geq 0$, and $\|A\| = 0$ if and only if $A = 0$
- 2 $\|A + B\| \leq \|A\| + \|B\|$
- 3 $\|\alpha A\| = |\alpha| \|A\|$

Example (Vec norm). Stack the columns of A into one long vector,

$$\text{vec}(A) := [a_{11} \ a_{21} \ \cdots \ a_{n1} \ a_{12} \ \cdots \ a_{nn}]^T \in \mathbb{C}^{n^2},$$

and set $\|A\| := \|\text{vec}(A)\|$ for any vector norm on $\mathbb{C}^{n^2}$.

Example (Frobenius norm).

$$\|A\|_F := \|\text{vec}(A)\|_2 = \sqrt{\sum_{j,k=1}^n |a_{jk}|^2}.$$

Induced (lub) Matrix Norms

Definition (Induced matrix norm). Let $\|\cdot\|$ be a vector norm on $\mathbb{C}^n$. For $A \in \mathbb{C}^{n \times n}$,

$$\text{lub}(A) = \text{lub}_{\|\cdot\|}(A) := \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \max_{\|x\|=1} \|Ax\| :$$

the largest factor by which A stretches a vector (lub = least upper bound).

Remark (Why this makes sense).

- The two maxima agree: stretching does not depend on the length of x , since $\|Ax\| / \|x\| = \|A(x/\|x\|)\|$ and $x/\|x\|$ has length 1.
- A largest value exists: $x \mapsto \|Ax\|$ is continuous, and a continuous function on the unit sphere $\{x : \|x\| = 1\}$ (closed and bounded) has a largest value.

Remark (Basic inequality). No vector is stretched more than the largest stretch: $\|Ax\| \leq \text{lub}(A) \|x\|$ for all $A \in \mathbb{C}^{n \times n}$, $x \in \mathbb{C}^n$ (for $x = 0$ both sides are 0).

lub Is a Matrix Norm

Proposition. For every vector norm $\|\cdot\|$ on $\mathbb{C}^n$, $\text{lub}_{\|\cdot\|}$ is a matrix norm. It also satisfies $\text{lub}(I) = 1$.

Proof. Idea: each property holds for every single vector x with $\|x\| = 1$; then it holds for the largest stretch too.

1. *Positive.* Stretches are ≥ 0 , so $\text{lub}(A) \geq 0$. If $\text{lub}(A) = 0$, then $Ax = 0$ for every x ; with $x = e_k$ (k -th unit vector), column k of A is 0, for each k . So $A = 0$.

2. *Triangle inequality.* For $\|x\| = 1$, by the vector triangle inequality,

$$\|(A + B)x\| \leq \|Ax\| + \|Bx\| \leq \text{lub}(A) + \text{lub}(B).$$

3. *Scaling.* $\|\alpha Ax\| = |\alpha| \|Ax\|$ for every x , so the largest values also differ by the factor $|\alpha|$.

4. *Identity.* I does not stretch: $\|Ix\| = \|x\| = 1$. ■

The Three Classical lub Norms

Proposition. For $A \in \mathbb{C}^{n \times n}$:

Norm on $\mathbb{C}^n$	Induced matrix norm
$\ \cdot\ _1$	$\ A\ _1 := \text{lub}_{\ \cdot\ _1}(A) = \max_{1 \leq k \leq n} \sum_{j=1}^n a_{jk} $ (max column sum)
$\ \cdot\ _\infty$	$\ A\ _\infty := \text{lub}_{\ \cdot\ _\infty}(A) = \max_{1 \leq j \leq n} \sum_{k=1}^n a_{jk} $ (max row sum)
$\ \cdot\ _2$	$\ A\ _2 := \text{lub}_{\ \cdot\ _2}(A) = \sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^H A)}$

Remark (Words). A^H = conjugate transpose. $\lambda_{\max}(A^H A)$ = largest eigenvalue of $A^H A$ (it is real and ≥ 0). $\sigma_{\max}(A)$ = largest singular value of A . Proofs: 1-norm and ∞ -norm below; 2-norm in Lectures 25–29.

Submultiplicativity

Theorem (*lub norms are submultiplicative*).

$$\text{lub}(AB) \leq \text{lub}(A) \text{lub}(B) \quad \text{for all } A, B \in \mathbb{C}^{n \times n}.$$

Proof. Idea: ABx means “first apply B , then A ”. B stretches by at most $\text{lub}(B)$, then A by at most $\text{lub}(A)$; stretch factors multiply.

1. *Two stretches.* Let $\|x\| = 1$. Apply the basic inequality to A (with the vector Bx), then to B (with the vector x):

$$\|ABx\| = \|A(Bx)\| \leq \text{lub}(A) \|Bx\| \leq \text{lub}(A) \text{lub}(B) \|x\| = \text{lub}(A) \text{lub}(B).$$

2. *Largest value.* This holds for every x with $\|x\| = 1$, hence also for the largest $\|ABx\|$, which is $\text{lub}(AB)$. ■

Remark. In particular $\|A^k\| \leq \|A\|^k$ for $k = 1, 2, \dots$ (induction on k : $\|A^{k+1}\| = \|AA^k\| \leq \|A\| \|A^k\|$).

Filling in the Details: The 1-Norm Formula

Proposition. $\|A\|_1 = \max_k \|a_k\|_1$, where $a_1, \dots, a_n$ are the columns of A (max column sum).

Proof. Idea: $Ax = \sum_k x_k a_k$ is a mix of the columns with weights whose sizes add up to 1. A mix cannot be bigger than the biggest column, and putting all the weight on the biggest column gives exactly that. Let $M := \max_k \|a_k\|_1$.

1. No x does better than M . Let $\sum_k |x_k| = 1$. By the triangle inequality,

$$\|Ax\|_1 = \left\| \sum_k x_k a_k \right\|_1 \leq \sum_k |x_k| \|a_k\|_1 \leq M \sum_k |x_k| = M.$$

2. One x reaches M . Let column k^* have $\|a_{k^*}\|_1 = M$ and take $x = e_{k^*}$ (so $\|x\|_1 = 1$): $Ax = a_{k^*}$ and $\|Ax\|_1 = M$. ■

Remark. Evaluating $\|A\|_1$ or $\|A\|_\infty$ costs about n^2 operations: cheap to compute in practice.

Filling in the Details: The ∞ -Norm Formula

Proposition. $\|A\|_\infty = \max_j \sum_k |a_{jk}|$ (max row sum).

Proof. Idea: entry j of Ax is row j times x , with every $|x_k| \leq 1$. It is largest when no terms cancel: choose each x_k to match the sign of a_{jk} . Let

$$M := \max_j \sum_k |a_{jk}|.$$

1. No x does better than M . Let $\|x\|_\infty = 1$, so every $|x_k| \leq 1$. For each row j ,

$$|(Ax)_j| = \left| \sum_k a_{jk} x_k \right| \leq \sum_k |a_{jk}| |x_k| \leq \sum_k |a_{jk}| \leq M.$$

2. One x reaches M . Let row j^* have $\sum_k |a_{j^*k}| = M$. Real A : take $x_k := \text{sgn}(a_{j^*k}) = \pm 1$ (and 1 if $a_{j^*k} = 0$). Complex A : take $x_k := \overline{a_{j^*k}} / |a_{j^*k}|$. Either way $\|x\|_\infty = 1$, $a_{j^*k} x_k = |a_{j^*k}|$, and

$$\|Ax\|_\infty \geq |(Ax)_{j^*}| = \sum_k |a_{j^*k}| = M.$$



Filling in the Details: The 2-Norm

Remark (No entrywise formula).

$$\|A\|_2 = \sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^H A)}$$

needs an eigenvalue computation (Lectures 25–29), not just sums of entries.

Proposition. If $U \in \mathbb{C}^{n \times n}$ is unitary ($U^H U = I$), then $\|Ux\|_2 = \|x\|_2$ for all x , and $\|U\|_2 = 1$.

Proof. Idea: a unitary matrix is a rotation or reflection; it keeps lengths.

1. *Lengths are kept.* Write the squared length as $v^H v$ and use $U^H U = I$:

$$\|Ux\|_2^2 = (Ux)^H (Ux) = x^H U^H U x = x^H x = \|x\|_2^2.$$

2. *Largest stretch.* Every x with $\|x\|_2 = 1$ has $\|Ux\|_2 = 1$, so the largest stretch is 1. ■

Which Norm When? (1/2)

Proposition. For all $x \in \mathbb{C}^n$:

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq n \|x\|_\infty.$$

Proof. 1. Largest entry $\leq$ length. Let $|x_m| = \|x\|_\infty$. One term of a sum of nonnegative terms is at most the sum: $|x_m|^2 \leq \sum_i |x_i|^2$. Take square roots.

2. Length $\leq$ total size. Square the total size; the cross terms are ≥ 0 :

$$\left(\sum_i |x_i|\right)^2 = \sum_i |x_i|^2 + \sum_{i \neq l} |x_i| |x_l| \geq \sum_i |x_i|^2.$$

3. Total size $\leq n \times$ largest entry. Each of the n terms of $\sum_i |x_i|$ is at most $\|x\|_\infty$.



Remark (All norms are comparable; proof omitted). For any two norms on $\mathbb{C}^n$ there are $c, c' > 0$ with $c \|x\|_a \leq \|x\|_b \leq c' \|x\|_a$ for all x . So “ $= O(\epsilon_{\text{mach}})$ ” or “converges” holds in one norm if and only if it holds in every norm.

Which Norm When? (2/2)

- **Constants do change.** $\kappa_1(A)$, $\kappa_2(A)$, $\kappa_\infty(A)$ are different numbers in general; they agree up to factors that depend only on n . Say which norm you use.
- $\|\cdot\|_1$, $\|\cdot\|_\infty$: easy to compute; good for bounds and in software.
- $\|\cdot\|_2$: unitarily invariant, $\|UAV\|_2 = \|A\|_2$ for unitary U, V (rotations and reflections do not change it). The natural norm for the eigenvalue part of the course.
- $\|\cdot\|_F$: also unitarily invariant and easy to compute, but for $n \geq 2$ it is *not* induced by any vector norm: $\|I\|_F = \sqrt{n} \neq 1$, while every lub norm has $\text{lub}(I) = 1$.

Looking Ahead

- **Next lecture:** Conditioning of $Ax = b$; Stability of LU Factorization